

The neural code

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Optimal Wiener Kernel

Consider the problem that from the history of the stimulus, we would like to estimate or predict the response of this sensory neuron $r_{est}(t)$. Let us consider the simplest functional model, namely

$$r_{est}(t) = r_0 + \int_0^\infty D(\tau)s(t-\tau)d\tau \quad (1)$$

r_0 is considered to be a constant. As we can see, if the functional derivative of r with respect to s does not depend on the choice of t , then

$$\frac{\delta r}{\delta s(t-\tau)} = D(\tau), \quad (2)$$

$D(\tau)$ is also called the first Wiener Kernel in the literature.

Now let us define an error function,

$$E = \int_0^T dt (r_{est}(t) - r(t))^2 \quad (3)$$

Our goal is to minimize the error function and find the optimal kernel.

Therefore, we have

$$\begin{aligned} \frac{\delta E}{\delta D} = 0 &= \int_0^T dt (r_{est}(t) - r(t)) \frac{\delta r_{est}}{\delta D}. \\ \frac{\delta r_{est}}{\delta D(\tau)} &= \int_0^T d\tau' s(t-\tau')\delta(\tau'-\tau) = s(t-\tau). \end{aligned} \quad (4)$$

As a result, we have

$$\int_0^T dt \int_0^\infty d\tau' D(\tau')s(t-\tau')s(t-\tau) = \int_0^T dt (r(t) - r_0)s(t-\tau) \quad (5)$$

Rearranging the integral on the left, we have

$$\int_0^\infty D(\tau') d\tau' \int_0^T dt s(t - \tau') s(t - \tau) = \int_0^T dt (r(t) - r_0) s(t - \tau) \quad (6)$$

Now, we can define the autocorrelation function of the stimulus as

$$Q_{ss}(\tau) = \frac{1}{T} \int_0^T s(t) s(t + \tau) dt, \quad (7)$$

We can also define the correlation function between stimulus and response as

$$Q_{rs}(\tau) = \frac{1}{T} \int_0^T r(t) s(t + \tau) dt. \quad (8)$$

Then,

$$\int_0^\infty D(\tau') d\tau' Q_{ss}(\tau - \tau') = Q_{rs}(-\tau) \quad (9)$$

Now consider the simplest case in which the stimuli is white noise. In this case $Q_{ss}(\tau) = \sigma^2 \delta(\tau)$. Then

$$\int_0^\infty D(\tau') \sigma^2 \delta(\tau - \tau') d\tau' = D(\tau) \sigma^2 \quad (10)$$

As a result, we find the optimal kernel has an explicit expression

$$D(\tau) = \frac{Q_{rs}(-\tau)}{\sigma^2} \quad (11)$$

When the stimuli is not white noise, it is hard to calculate the kernel explicitly. However, if we ignore the causality, and make the assumption that response not only depends on the past but also the future of the stimulus, then things become much easier. performing Fourier transform, we found that in the frequency domain

$$\tilde{D}(\omega) \tilde{Q}_{ss}(\omega) = \tilde{Q}_{rs}(-\omega) \quad (12)$$

The Spike-Triggered Average

Now let me introduce the concept of spike triggered average, which is defined as

$$C(\tau) = \left\langle \frac{1}{n} \sum_{i=1}^n s(t_i - \tau) \right\rangle \quad (13)$$

Here t_i is the time when a spike occurs. The average $\langle \dots \rangle$ is over different trials during which we are presenting exactly the same stimulus. The timing of the spike, however, can be variable. Below I would like to show that the spike triggered average has a direct connection the stimulus response correlation function $Q_{rs}(-\tau)$, provided that r is defined as the firing rate of a neuron. Given the density function

$$\rho(t) = \sum_i \delta(t - t_i) \quad (14)$$

It is easy to see that the

$$\sum_{i=1}^n s(t_i - \tau) = \int_0^T \rho(t) s(t - \tau) dt$$

Thus,

$$C(\tau) = \left\langle \frac{1}{n} \int_0^T \rho(t) s(t - \tau) dt \right\rangle$$

Formally, the firing rate of a neuron $r(t)$ is defined as

$$r(t) = \frac{1}{\Delta t} \int_t^{t+\Delta t} d\tau \langle \rho(\tau) \rangle, \quad (15)$$

where the average $\langle \dots \rangle$ is also over different trials. Next, we want to make an important approximation by replacing the number of spikes n within a single trial with the mean number of spikes across all trials $\langle n \rangle$ during the period T . Then, because we are using exactly the same stimulus on each trial, the spike triggered average can be reduced to

$$\begin{aligned} C(\tau) &\approx \frac{1}{\langle n \rangle} \int_0^T \langle \rho(t) \rangle s(t - \tau) dt \\ &= \frac{1}{\langle n \rangle} \int_0^T r(t) s(t - \tau) dt \\ &= \frac{T}{\langle n \rangle} Q_{rs}(-\tau) \\ &= \frac{1}{\langle r \rangle} Q_{rs}(-\tau) \end{aligned} \quad (16)$$

Linear nonlinear model

The optimal kernel derived from a linear model has two problems. First, there is nothing to prevent the predicted response to become negative, and the predicted response does not saturate. As the magnitude of the response increases, the response would also increase without bound. If we use L to represent the linear term we have been discussing thus far:

$$L(t) = \int_0^\infty d\tau D(\tau) s(t - \tau) \quad (17)$$

The modification is to replace the linear prediction $r_{est}(t) = r_0 + L(t)$ with the generalization

$$r_{est}(t) = r_0 + F[L(t)] \quad (18)$$

Reverse-Correlation Methods and the Receptive Field of a Simple Cell

With the above preparations, we are now at a position to map the receptive field a sensory neuron in the visual cortex. The spike-triggered average for visual stimuli is defined, as the average over trials of stimuli evaluated at times $t_i - \tau$, where t_i is the spike times. Because the light intensity of a visual image depends on locations as well as time, the spike-triggered average stimulus is a function of three variables,

$$C(x, y, \tau) \approx \frac{1}{\langle n \rangle} \left\langle \sum_{i=1}^n s(x, y, t_i - \tau) \right\rangle \quad (19)$$

The visual stimulus we are presenting is Gaussian white noise that is uncorrelated in both space and time,

$$\langle s(x, y, t) s(x', y', t') \rangle = \sigma_s^2 \delta(t - t') \delta(x - x') \delta(y - y') \quad (20)$$

We thus have

$$L(t) = \int_0^\infty d\tau \int dx dy D(x, y, \tau) s(x, y, t - \tau), \quad (21)$$

where the optimal kernel can be found through spike-triggered average

$$D(x, y, \tau) = \frac{\langle r \rangle C(x, y, \tau)}{\sigma_s^2} \quad (22)$$

The kernel $D(x, y, \tau)$ defines the space-time receptive field of a neuron. For some neurons, the kernel can be written as a product of two functions, one that describes the spatial and the other describes the temporal receptive field, namely

$$D(x, y, \tau) = D_s(x, y) D_t(\tau) \quad (23)$$

And the linear response of a neuron can also be viewed as a product of two parts

$$L(t) = L_s L_t \quad (24)$$

To derive and understand the spatial filter, let us consider an array of retinal ganglion cells with receptive field covering a small patch of the retina. We assume that the statistics of the visual inputs on that small patch are stationary and translational invariant, namely

$$\langle s(x) \rangle = \langle s(x + u) \rangle \quad (25)$$

$$\langle s(x) s(x + u) \rangle = f(u) \quad (26)$$

This means all locations and directions (as well as all times) in that patch are the same, which is equivalent for all the neurons in that patch to have the same receptive field or kernel. Under this assumption, the response of the neuron that localizes at (x_0, y_0) can be viewed as the convolution of the kernel with the visual stimulus, namely

$$L_s(x_0, y_0) = \int \int dx dy D_s(x - x_0, y - y_0) s(x, y) \quad (27)$$

$$L_t(t) = \int_0^\infty d\tau D_t(\tau) s(t - \tau) \quad (28)$$

The above network now has a very sexy name, CNN.

Shannon Information

Shannon information can be formulated intuitively by playing the game “sixty-three”. If you have a number in your hand, and this integer is randomly picked from 0 and 63 with equal probability. What’s the smallest number of yes/no questions needed to identify an integer x between 0 and 63?

Intuitively, Six questions suffice. One reasonable strategy ask the following questions:

- $x \geq 32$?
- is $x \bmod 32 \geq 16$?
- is $x \bmod 16 \geq 8$?
- is $x \bmod 8 \geq 4$?
- is $x \bmod 4 \geq 2$?
- is $x \bmod 2 \geq 1$?

What are the Shannon information contents of the outcomes in this example? If we assume that all values of x are equally likely, then the answers to the questions are independent and each has Shannon information content $\log_2(1/0.5) = 1\text{bit}$; the total Shannon information gained is always $\log_2 64 = 6$ bits. Furthermore, the number x that we learn from these questions is a six-bit binary number. Our questioning strategy defines a way of encoding the random variable x as a binary file.

Now let us put everything in the context of sensory encoding. Given a sensory stimulus, such as flash light with given intensity, neurons in the retina, such as bipolar cells, can faithfully respond to the intensity change. Let us denote the response amplitude r , which could be membrane potential, or the firing rate of the neuron. Just by measuring the response of the neuron, how

much information we could learn about the stimulus. The answer is just given by

$$h = -\log_2 P(r)$$

where $P(r)$ is the probability of a given response amplitude.

When we write down the above formula, we are assuming that all responses have equal probability. When responses have unequal probabilities, we modify the above formula, by computing the average over all different responses. This leads to the definition of Shannon information

$$I = -\sum_r P(r) \log_2 P(r) \quad (29)$$

When we treat the response of a neuron as a continuous variable, one should then define the probability density function $p(r)$, and the Shannon information becomes

$$I = -\sum_r p(r) \Delta r \log_2 p(r) \Delta r \quad (30)$$

Furthermore,

$$\lim_{\Delta r \rightarrow 0} (I + \log_2 \Delta r) = -\int dr p(r) \log_2 p(r) \quad (31)$$

We now ask, what is the input-output relationship of a neuron, $r = f(s)$, that would maximize the information (or entropy) of the response? Let's consider the simple case, by assuming that there is a maximum response r_{max} a neuron could achieve. This could be the maximum firing rate. And the minimum response is zero. It is now your homework to show that under this condition, different responses with equal probability would maximize the entropy. Now use the normalization condition

$$\int_0^{r_{max}} p(r) dr = 1,$$

we found that $p(r) = \frac{1}{r_{max}}$. Given the probability density distribution of inputs $p(s)$, One must have $p(s) ds = p(r) dr$. As a result, $\frac{dr}{ds} = r_{max} p(s)$, and the input-output function should be proportional to the cumulative probability distribution of stimulus

$$f(s) = \int_0^s r_{max} p(s') ds' \quad (32)$$

Law of diminished returns

Let us consider another scenario such that the mean firing rate of a neuron is fixed. This could be due to the fact that transmitting information with higher firing rate requires greater metabolic cost, and the cost E is a monotonic

increasing function of the firing rate. In the simplest case, let us discretize the activity of a neuron into multiple levels, as we did earlier. Maximizing the information for a fixed mean cost is reduced to maximize the following objective function

$$G = - \sum_n P_n \ln P_n + \beta \left(\sum_n P_n E_n - E \right) + \lambda \left(\sum_n p_n - 1 \right) \quad (33)$$

This yields

$$P_n = \frac{1}{Z} \exp(-\beta E_n) \quad (34)$$

The famous Boltzmann distribution where

$$Z = \sum_n \exp(-\beta E_n) \quad (35)$$

is the famous partition function in statistical physics, and β is the inverse temperature. The mean cost or energy

$$E = \frac{\sum_n \exp(-\beta E_n) E_n}{Z} = - \frac{\partial \ln Z}{\partial \beta} \quad (36)$$

One can also examine the function $H(E)$, namely how the entropy or information changes as a function of energy or cost. At zero temperature, the entropy would be zero. At very high temperature, all levels would be equally likely, and the entropy or information would be $H = \ln N$. The H would be a concave function of E , namely if you pay the cost $E_m = (E_1 + E_2)/2$, we can transmit at least rate $(H_1 + H_2)/2$ by dividing your time equally between the 1 and 2 codes.

Now, what happened if we would like to maximize the information per cost, namely H/E . Again, each coding level will be described by the Boltzmann distribution. And the question is what is the inverse temperature which would maximize H/E ? Recall the identity you learned statistical physics (this could be a homework to check),

$$H = \beta E + \ln Z \quad (37)$$

By taking the derivative $\frac{\partial(H/E)}{\partial \beta} = 0$, we find the following condition

$$\ln Z \frac{\partial \ln^2 Z}{\partial \beta^2} = 0$$

We therefore arrives at a very interesting result,

$$Z = 1 \quad (38)$$

is when the information rate per cost is maximized.

Mutual Information

As we discussed in previous lectures, the entropy of the neural response is given by

$$H_r = - \sum_r P(r) \log_2 P(r) \quad (39)$$

In early sensory processing, what we are really interested is how much useful information has been transmitted from the input to the output. Are all the bits in H_r useful? Not really, some of the bits may reflect noise. Given the stimulus value s , the response can be variable, as defined by the conditional probability $P(r|s)$. Thus the noise entropy is given by

$$H_{noise} = - \sum_r P(r|s) \log_2 P(r|s)$$

When we averaged this over different stimulus, one has

$$\langle H_{noise} \rangle = - \sum_s \sum_r P(s) P(r|s) \log_2 P(r|s) \quad (40)$$

The useful information is the subtraction of noise entropy from the response entropy, called mutual information

$$\begin{aligned} I &= H_r - H_{noise} \\ &= - \sum_r P(r) \log_2 P(r) + \sum_s \sum_r P(s) P(r|s) \log_2 P(r|s) \\ &= - \sum_r \sum_s P(r|s) P(s) \log_2 P(r) + \sum_s \sum_r P(s) P(r|s) \log_2 P(r|s) \\ &= \sum_r \sum_s P(r|s) P(s) \log_2 \frac{P(r|s)}{P(r)} \end{aligned} \quad (41)$$

Recall that joint probability

$$P(r, s) = P(r|s)P(s),$$

the mutual information can then be expressed in a symmetric form between stimulus and response,

$$I = \sum_r \sum_s P(r, s) \log_2 \frac{P(r, s)}{P(r)P(s)} \quad (42)$$

Mutual information also provides a distance measure between two distributions, called K-L divergence, formally defined as

$$D_{KL}(P, Q) = \sum_x P(x) \log_2 \frac{P(x)}{Q(x)} \quad (43)$$

It is your homework to show that $D_{KL} \geq 0$ and $D_{KL} = 0$ only when $P = Q$.

Mutual information of Gaussian variable

Consider a gaussian random variable x drawn from the distribution

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{(x - \langle x \rangle)^2}{2\sigma^2} \right] \quad (44)$$

Let us compute its entropy

$$S = - \int_{-\infty}^{\infty} p(x) \log_2 p(x) \quad (45)$$

To compute it explicitly, we have

$$S = \frac{1}{\ln 2} \left[\ln \sqrt{2\pi\sigma^2} + \int p(x) \frac{(x - \langle x \rangle)^2}{2\sigma^2} \right] \quad (46)$$

$$= \frac{1}{\ln 2} \left[\frac{1}{2} \ln(2\pi\sigma^2) + \frac{1}{2} \right] \quad (47)$$

$$= \frac{1}{\ln 2} \frac{1}{2} [\ln(2\pi\sigma^2) + 1] \quad (48)$$

$$= \frac{1}{\ln 2} \frac{1}{2} \ln(2\pi e\sigma^2) \quad (49)$$

$$= \frac{1}{2} \log_2(2\pi e\sigma^2) \quad (50)$$

Now let us consider a very simple linear model

$$y = wx + \xi \quad (51)$$

Here ξ is noise with mean $\langle \xi \rangle$ zero and variance σ_ξ^2 .

$$I(x, y) = S(y) - \langle S(y|x) \rangle_x \quad (52)$$

$$= S(y) - \langle S(wx + \xi|x) \rangle_x \quad (53)$$

$$= S(y) - \langle S(\xi|x) \rangle_x \quad (54)$$

$$= S(y) - S(\xi) \quad (55)$$

$$= \frac{1}{2} \log_2 \frac{\sigma_y^2}{\sigma_\xi^2} \quad (56)$$

$$= \log_2 \frac{\sigma_y}{\sigma_\xi} \quad (57)$$

In addition, we know that

$$\sigma_y^2 = w^2 \sigma_x^2 + \sigma_\xi^2 \quad (58)$$

Therefore,

$$I(x, y) = \frac{1}{2} \log_2 \left(1 + \frac{w^2 \sigma_x^2}{\sigma_\xi^2} \right) \quad (59)$$

Since we can rewrite

$$y = w(x + \xi_{eff}) \quad (60)$$

where $\xi_{eff} = \xi/w$. We thus have $\langle \xi_{eff}^2 \rangle = \sigma_\xi^2/w^2$. Define the signal to noise ratio as

$$\mathbf{SNR} = \frac{\langle x^2 \rangle}{\langle \xi_{eff}^2 \rangle} \quad (61)$$

The mutual information

$$I(x, y) = \frac{1}{2} \log_2(1 + \mathbf{SNR}) \quad (62)$$

We can also generalize the above case to multivariate Gaussian distribution. Consider the model

$$\mathbf{y} = \mathbf{W}\mathbf{x} + \xi \quad (63)$$

Let us define the covariance matrices for the output and the noise as

$$\mathbf{C}_Y = \langle \mathbf{y}\mathbf{y}^T \rangle \quad (64)$$

$$\mathbf{C}_\xi = \langle \xi\xi^T \rangle \quad (65)$$

The mutual information is now given by

$$I(x, y) = \frac{1}{2} \log_2 \frac{|\mathbf{C}_Y|}{|\mathbf{C}_\xi|}, \quad (66)$$

where $|\mathbf{C}|$ is the determinant of the covariance matrix.

Sensory encoding in multiple neurons

With all the above preparations, we could now consider a very interesting feed-forward model. The output response in neuron y_i is given by

$$y_i = \sum_j W_{ij}(x_j + z_j) + \eta_i \quad (67)$$

Here x may be viewed as the voltage signal from the photoreceptor, and we are measuring the response in the bipolar cell in the retina. $\mathbf{x} = [x_1, x_2, \dots, x_N]^T$ with zero means drawn from a multivariate gaussian distribution

$$p(\mathbf{x}) = \frac{1}{Z} \exp\left(-\frac{1}{2}\mathbf{x}^T \mathbf{C}^{-1} \mathbf{x}\right) \quad (68)$$

where Z is the normalization factor. $\mathbf{C}$ is the covariance matrix of $\mathbf{x}$, satisfying $\mathbf{C} = \langle \mathbf{x}\mathbf{x}^T \rangle$. η_i is noise drawn also from a gaussian distribution, whose covariance matrix is given by $\langle \eta_i \eta_j \rangle = \sigma_\eta^2 \delta_{ij}$. z_j is the input noise, and $\langle z_i z_j \rangle = \sigma_z^2 \delta_{ij}$.

By defining a new effective noise

$$\xi_i = \sum_j W_{ij} z_j + \eta_i, \quad (69)$$

we can now use the formula above, which gives us

$$I = \frac{1}{2} \log_2 \frac{|\mathbf{C}_Y|}{|\mathbf{C}_\xi|}. \quad (70)$$

Maximizing the mutual information may be achieved by boosting the signal. However, there is an energy cost. Here let us set a constraint on the total output signal, that is

$$\sum_i \langle y_i^2 \rangle = \text{Tr} \langle \mathbf{y} \mathbf{y}^T \rangle \quad (71)$$

Thus our job is to maximize the following objective function

$$F = \frac{1}{2} \log_2 \frac{|\mathbf{C}_Y|}{|\mathbf{C}_\xi|} - \lambda \text{Tr} \langle \mathbf{y} \mathbf{y}^T \rangle \quad (72)$$

Writing it down explicitly, we have

$$\mathbf{C}_Y = \langle (\mathbf{W}(\mathbf{x} + \mathbf{z}) + \eta)(\mathbf{W}(\mathbf{x} + \mathbf{z}) + \eta)^T \rangle \quad (73)$$

$$= \langle \mathbf{W} \mathbf{x} \mathbf{x}^T \mathbf{W}^T \rangle + \langle \mathbf{W} \mathbf{z} \mathbf{z}^T \mathbf{W}^T \rangle + \langle \eta \eta^T \rangle \quad (74)$$

Likewise,

$$\mathbf{C}_\xi = \langle \mathbf{W} \mathbf{z} \mathbf{z}^T \mathbf{W}^T \rangle + \langle \eta \eta^T \rangle \quad (75)$$

Now let us consider choosing an orthogonal matrix $\mathbf{U} \mathbf{U}^T = \mathbf{U}^T \mathbf{U} = \mathbf{I}$ that can diagonalize the input and output covariance matrix (see the translational invariant covariance spectrum as discussed in the next section)

$$\mathbf{U} \langle \mathbf{x} \mathbf{x}^T \rangle \mathbf{U}^T = \begin{bmatrix} \diagdown & & \\ & c(k) & \\ & & \diagdown \end{bmatrix} \quad (76)$$

Since both determinant and trace are self-invariant under a similarity transformation, we can take

$$\mathbf{U} \mathbf{C}_Y \mathbf{U}^T = \hat{\mathbf{W}} \begin{bmatrix} \diagdown & & \\ & c(k) & \\ & & \diagdown \end{bmatrix} \hat{\mathbf{W}}^T + \hat{\mathbf{W}} \begin{bmatrix} \diagdown & & \\ & \sigma_z^2 & \\ & & \diagdown \end{bmatrix} \hat{\mathbf{W}}^T + \mathbf{U} \begin{bmatrix} \diagdown & & \\ & \sigma_\eta^2 & \\ & & \diagdown \end{bmatrix} \mathbf{U}^T, \quad (77)$$

where

$$\hat{\mathbf{W}} = \mathbf{U} \mathbf{W} \mathbf{U}^T = \begin{bmatrix} \diagdown & & \\ & w(k) & \\ & & \diagdown \end{bmatrix} \quad (78)$$

The objective function can now be expressed as

$$F = \frac{1}{2} \sum_i \left[\log_2 \frac{c(k)w^2(k) + \sigma_z^2 w^2(k) + \sigma_\eta^2}{\sigma_z^2 w^2(k) + \sigma_\eta^2} - 2\lambda (c(k)w^2(k) + \sigma_z^2 w^2(k) + \sigma_\eta^2) \right] \quad (79)$$

Let us first consider a special case $\sigma_z^2 = 0$ Then

$$F = \frac{1}{2} \sum_i \left[\log_2 \frac{c(k)w^2(k) + \sigma_\eta^2}{\sigma_\eta^2} - 2\lambda (c(k)w^2(k) + \sigma_\eta^2) \right] \quad (80)$$

By taking $\frac{\partial F}{\partial w^2(k)} = 0$, we have

$$\begin{aligned} \frac{1}{1 + \frac{c(k)w^2(k)}{\sigma_\eta^2}} &= 2\lambda \\ \text{or } c(k)w^2(k) &= \text{const} \end{aligned} \quad (81)$$

Since the eigenvalue of the output covariance matrix is $c(k)w^2(k) + \sigma_\eta^2$, efficient coding principle basically says that one would make all the eigenvalues equal, i.e., $\mathbf{C}_Y \sim \mathbf{I}$. This means efficient coding prefers decorrelation and whitening!

Now consider a more general case $\sigma_z^2 \neq 0$. By defining $\mathbf{SNR} = \hat{c}(k) = \frac{c(k)}{\sigma_z^2}$, $\hat{w}^2(k) = \frac{w^2(k)\sigma_z^2}{\sigma_\eta^2}$, $\hat{\lambda} = 2\lambda\sigma_\eta^2$, the objective function can be rewritten as

$$F = \frac{1}{2} \sum_i \left[\log_2 \frac{\hat{c}(k)\hat{w}^2(k) + \hat{w}^2(k) + 1}{\hat{w}^2(k) + 1} - \hat{\lambda} (\hat{c}(k)\hat{w}^2(k) + \hat{w}^2(k) + 1) \right] \quad (82)$$

When the signal to noise level is small $\hat{c}(k) \ll 1$, F is maximized when

$$\begin{aligned} \frac{\hat{c}(k)}{(\hat{w}^2(k) + 1)^2} &= \hat{\lambda} \\ \text{or } \hat{w}^2(k) &= 0 \end{aligned} \quad (83)$$

In the opposite limit $\hat{c}(k) \gg 1$, F is maximized when

$$\hat{w}^2(k) \sim \frac{1}{\hat{c}(k)} \quad (84)$$

Stationary statistics and optimal coding

An important feature of many important natural stimuli is their stationarity. Consider grey level images with n pixels, the components of the input vector, x_i , which corresponds to the grey level light intensity (or contrast) of the image pixel indexed by i . Furthermore, since we are not interested in mean pixel values it is convenient to define the x_i to refer to the image pixel value relative to its mean. Natural images are well known to be redundant: obviously, neighboring pixels are highly correlated.

Spatial stationarity of the statistics of natural images means that the statistical properties are invariant to arbitrary global translations of pixels within the image. In other words, correlations between pixel values depend only on distances between the pixels, not on their absolute location. This is a very good approximation if we ignore the pixels at the boundaries of the image. A useful mathematical trick to get rid of boundary effects is impose a periodic boundary condition: to imagine that the image is not on a two dimensional flat grid but on a torus, so that there are literally no boundaries to worry about. So we have

$$C_{ij} = \langle x_i x_j \rangle = f(|\mathbf{r}_i - \mathbf{r}_j|) \quad (85)$$

A matrix with this property is called the circulant matrix

$$C = \begin{bmatrix} c_0 & c_1 & \dots & c_{n-2} & c_{n-1} \\ c_{n-1} & c_0 & c_1 & & \\ \vdots & c_{n-1} & c_0 & \ddots & \vdots \\ c_2 & & \ddots & \ddots & c_1 \\ c_1 & c_2 & \dots & c_{n-1} & c_0 \end{bmatrix}.$$

Moreover, our covariance matrix is symmetric, meaning that $c_{n-i} = c_i$, and the covariant matrix is determined by $\lfloor n/2 \rfloor + 1$ elements. One important property of the circulant matrix is that its eigenvectors are Fourier basis

$$\begin{aligned} \mathbf{u}_m &= [1, \omega_m, \omega_m^2, \dots, \omega_m^{n-1}]^T \\ \omega_m &= \exp\left(\frac{i2\pi m}{n}\right) \\ m &= 0, 1, \dots, n-1 \end{aligned} \quad (86)$$

And the corresponding eigenvalue is the discrete Fourier transform of the matrix elements c_i .

$$\begin{aligned} \lambda_m &= c_0 + c_1 \omega_m + c_2 \omega_m^2 + \dots + c_{n-1} \omega_m^{n-1} = \sum_{j=0}^{n-1} c_j \exp(ikj) \\ k &= \frac{2\pi m}{n} \end{aligned} \quad (87)$$

Now go back to the vision problem. Let us imagine the image to be on a square lattice with $L \times L$ pixels. Then the Fourier basis vectors $\mathbf{u}$ are,

$$\begin{aligned} u(k)_{ij} &= \exp(i\mathbf{k} \cdot \mathbf{R}_{ij}) \\ \mathbf{k} &= [k_x, k_y], \mathbf{r} = [r_x, r_y] \\ k_x, k_y &= \frac{2\pi m_{x,y}}{L}, m_{x,y} = 0, \pm 1, \dots, \pm \frac{L-1}{2} \\ r_x, r_y &= 0, \pm 1, \dots, \pm \frac{L-1}{2} \end{aligned} \quad (88)$$

The eigenvalue of the input covariance matrix $\mathbf{C}_x$ is now given by

$$\lambda(k) = \sum_{j=1}^{L^2} c_{1j} \exp(i\mathbf{k} \cdot \mathbf{R}_{1j}) = \frac{1}{L^2} \sum_i \sum_j c_{ij} \exp(i\mathbf{k} \cdot \mathbf{R}_{ij}) \quad (89)$$

Moreover, the eigenvalue of the covariant matrix is now, by definition the power spectra density of the image. To see this, we note that

$$\begin{aligned} c(k) &= \sum_i \sum_j c_{ij} \exp(i\mathbf{k} \cdot \mathbf{R}_{ij}) \\ &= \sum_i \sum_j \langle x(\mathbf{r}_i) x(\mathbf{r}_j) \rangle \exp(i\mathbf{k} \cdot (\mathbf{r}_i - \mathbf{r}_j)) \\ &= \langle \sum_i x(\mathbf{r}_i) \exp(i\mathbf{k} \cdot \mathbf{r}_i) \sum_j x(\mathbf{r}_j) \exp(-i\mathbf{k} \cdot \mathbf{r}_j) \rangle \\ &= \langle |\hat{x}(k)|^2 \rangle \end{aligned}$$

We shall now go back to our original optimization problem. The synaptic weight matrix $\mathbf{W}$ can be viewed as the receptive field of the output neurons $\mathbf{y}$. Let us assume that all the output neurons have the same receptive field. And the only difference is that the receptive field are centered at different positions of the visual field $\mathbf{r}_i$. In other words, we shall assume the weight matrix is also translational invariant, that is $W_{ij} = w(|\mathbf{r}_i - \mathbf{r}_j|)$. Therefore, the $\mathbf{W}$ matrix is also a symmetric circulant matrix, and we can use the same Fourier basis to diagonalize $\mathbf{W}$ and $\mathbf{C}_x$.

As a result, the efficient coding principle would predict, under certain assumptions, that in the limit of high signal to noise level $\hat{c}(k) = \frac{c(k)}{\sigma_z^2} \gg 1$

$$\hat{w}(k) \sim \frac{1}{\sqrt{\hat{c}(k)}} \quad (90)$$

In the limit of high noise level, $d^2 \ll 1$,

$$\hat{w}^2(k) = \sqrt{\frac{\hat{c}(k)}{\hat{\lambda}}} - 1 \quad (91)$$

So depending on the tradeoff parameter λ , either $\hat{w}(k) = 0$, or

$$\hat{w}(k) \sim \hat{c}^{1/4}(k) \sim \frac{c^{1/4}(k)}{\sigma_z^{1/2}} \quad (92)$$

For natural images, it is well known that the power spectra density decays

approximately as a power law, $\hat{c}(k) \sim \frac{1}{k^2}$, Thus we have that in the high SNR limit,

$$\begin{aligned}\hat{w}(k) &\sim k, \text{SNR} \gg 1 \\ \hat{w}(k) &\sim k^{-1/2}, \text{SNR} \ll 1\end{aligned}\tag{93}$$

Compare with experimental data...

These arguments - removing spatial correlations - can also be applied to the time domain. A time-dependent- signal has a power spectra density $S(\omega)$, and we can also ask what kind of temporal filters $K(\omega)$ would be optimal. Many naturally occurring signals have $1/f$ power spectra, this means that the optimal filter would obey $K^2(\omega) \sim \omega$ in the large SNR limit. When $\text{SNR} \ll 1$, we shall have $K^2(\omega) \sim 1/\sqrt{\omega}$.

These results can also be compared with experimental data...